

Aditya Dasika

AI/ML Engineer · Agentic & Multi-Agent Systems · RAG · LLM Fine-Tuning, Serving, Eval & Guardrails

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EXPERIENCE

AI/LLM Engineer II

April 2025 – Present

Phenom People

Hyderabad, Telangana

- Architected **chatbotCXA**gent, a production multi-agent system on **LangGraph** (12 orchestrated nodes) spanning intent recognition, planning, dialogue-state management and tool routing — with **Graph RAG** retrieval over enterprise knowledge and document repositories via FAISS vector search and a Neo4j graph for multi-hop, policy-aware grounding.
- Engineered tool-calling agents that execute governed actions — database queries and structured extraction from unstructured documents — under **XGrammar constrained decoding** and schema-validated JSON guardrails, raising tool-routing accuracy to **91.9%** and eliminating malformed tool calls in live traffic.
- Built an **LLM evaluation framework** (pass@k, LLM-as-judge rubric) gating every release across all 12 nodes, instrumented with **LiteLLM + Langfuse** for per-node cost, latency and full-trace auditability — continuously monitoring agent behaviour and curbing hallucinations.
- Fine-tuned and served Qwen models via **QLoRA** (Unsloth, NF4) on **vLLM** / AWS g5.xlarge (A10G), sustaining 10.98 RPS at 98.6% success under load at roughly half the inference cost of hosted GPT-4.1, and held the deployment to a golden baseline of 1.0s median and **1.4s** P95 latency with zero failures across **5,510** requests; separately fine-tuned a multi-task **ModernBERT** (intent + NER) in PyTorch, raising NLU accuracy from 87% to 97%.
- Building the detection and decisioning layer of an enterprise **hiring-integrity / fraud-risk platform** — an **LLM orchestration engine** that investigates flagged candidate activity, correlates evidence across signals and produces reviewer-facing assessments with supporting rationale — backed by a **policy-driven rules engine** and audited review workflows.
- Built the platform underneath it — a **FastAPI** on **MongoDB** configuration service with a versioned signal registry, resolved by **Temporal** workflows at workflow start in place of hardcoded per-tenant flags, and the self-hosted inference service as **Helm** releases with **OpenTelemetry** instrumentation feeding metrics dashboards; raised one service's test coverage from **63% to over 90%**.

Full Stack Engineer

May 2022 – February 2025

Blueleaves Farms

Hyderabad, Telangana

- Sole engineer** on the whole stack — owned end-to-end delivery of production systems (POS, CRM, inventory, procurement, e-commerce) as **React** and **Vite** front ends on Node.js, Firebase Cloud Functions and REST APIs, with role-based access control and CI/CD, cutting manual operational errors by roughly 90%.
- Built data-driven analytics on large-scale transactional and stock-movement data — demand forecasting, sales-trend and inventory models that surfaced discrepancies and improved procurement and replenishment decisions across 5+ locations and 10,000+ monthly transactions.

PROJECTS

IndexNotes — AI Knowledge Graph for Scientific Papers | *Neo4j, Ollama, PyMuPDF, Python, Docker*

- Built an automated Graph RAG pipeline ingesting research papers (PDF → local-LLM extraction → structured JSON → Neo4j), creating 40+ concepts and 17+ typed relationships, with quantized local models via Ollama.

NorthStar — Motorcycle Navigation Companion App | *Android, Kotlin, BLE, MapLibre*

- Reverse-engineered the Royal Enfield Tripper Dash's undocumented wire protocol — an **RSA-1024** plus **AES-256** handshake over UDP with auto-discovery, and an **H.264** over RTP video plane — then rendered navigation **off-screen** with **MapLibre** and **OSRM** so the phone's screen could stay off entirely, where the official app mirrors the display.

TECHNICAL SKILLS

GenAI & Agentic: Agentic / Multi-Agent Systems, LangGraph, LangChain, Tool/Function Calling, RAG & Graph RAG, Model Routing, Context-Window Management, Prompt Engineering, LLM Fine-Tuning (QLoRA, SFT)

Eval, Observability & Guardrails: LLM Evaluation (LLM-as-judge, pass@k), Constrained Decoding (XGrammar), Schema Validation, Langfuse, LiteLLM, OpenTelemetry, Model Monitoring, Hallucination Mitigation

ML & Serving: PyTorch, Transformers (BERT/ModernBERT), HuggingFace PEFT, Unsloth, vLLM, Embeddings / Vector Search (FAISS), Ollama, Neo4j, Production MLOps

Backend & Infra: Python, FastAPI, Node.js, REST APIs, Microservices, Docker, Helm, Temporal, AWS (EC2/S3/SageMaker), CI/CD, Linux

Databases: PostgreSQL, MongoDB, Neo4j, ClickHouse, Redis, Firebase

EDUCATION

Andhra Loyola Institute of Engineering and Technology

Vijayawada, Andhra Pradesh

Bachelor of Technology in Mechanical Engineering

August 2018 – May 2022